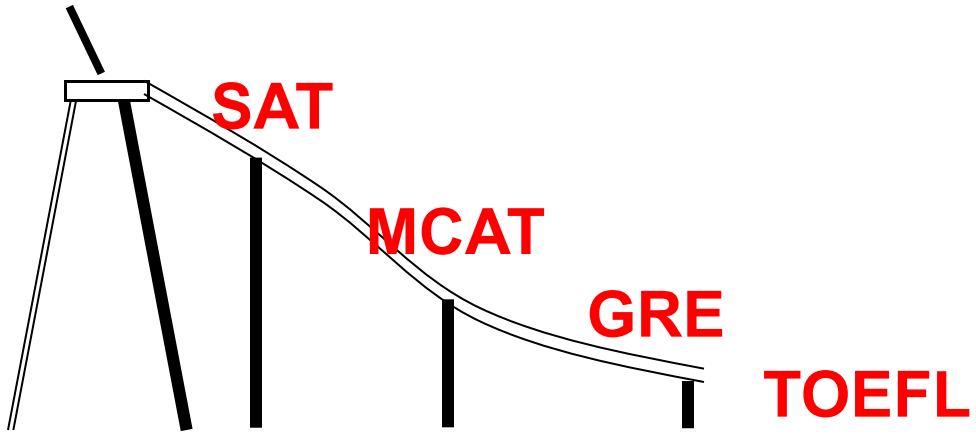
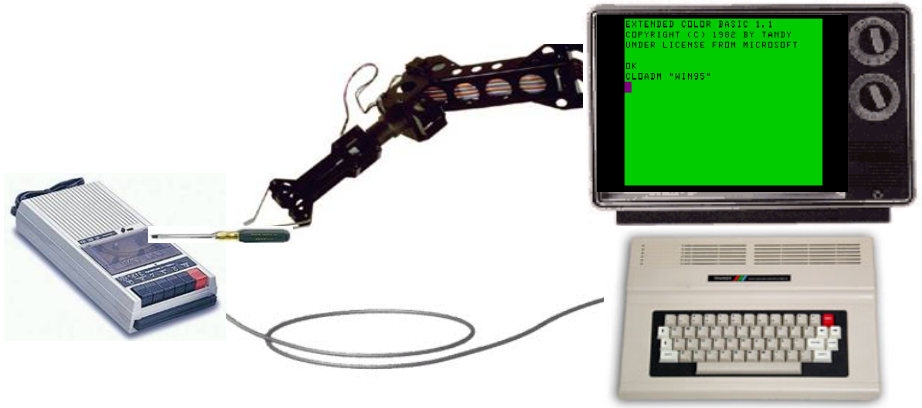
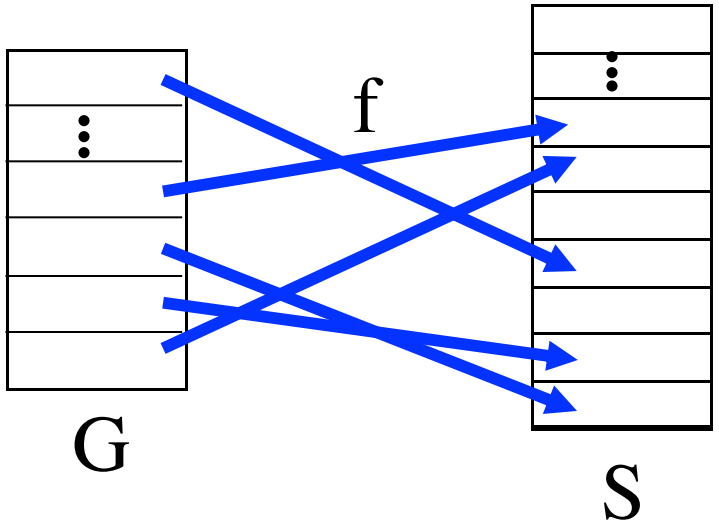
Test Slide



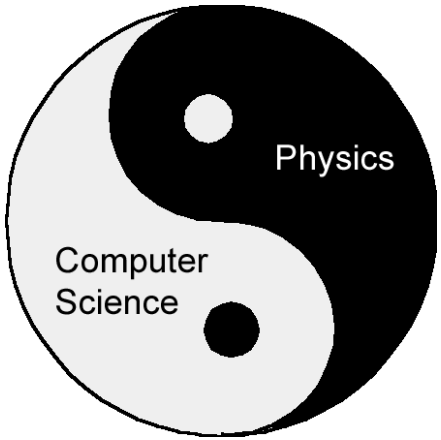
Quantum Computing in 2020



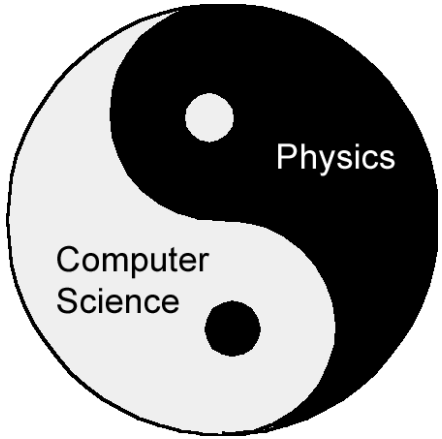


Dave Bacon

Santa Fe





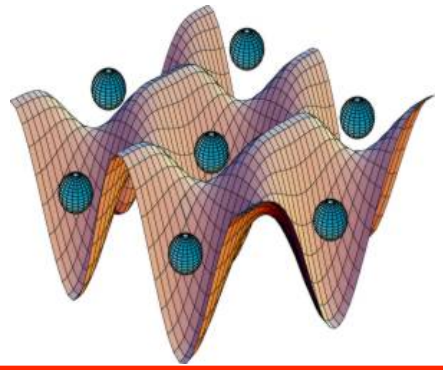


Apologies


MAPQUEST
Galice
66
61
Grants Pass
234
White City
Butte Falls
Willow Lake
Winema Natl. For.
0
10mi
10km
SINCE 1891
California
QUBITS
SESQUICENTENNIAL - 150 YEARS
CALTECH
Cecilville
Siskiyou
Trinity
Shasta
Dana
Big Bend
Trinity Center
©2003 MapQuest.com, Inc.

20th Century Reduced Rec





Quantum Computation



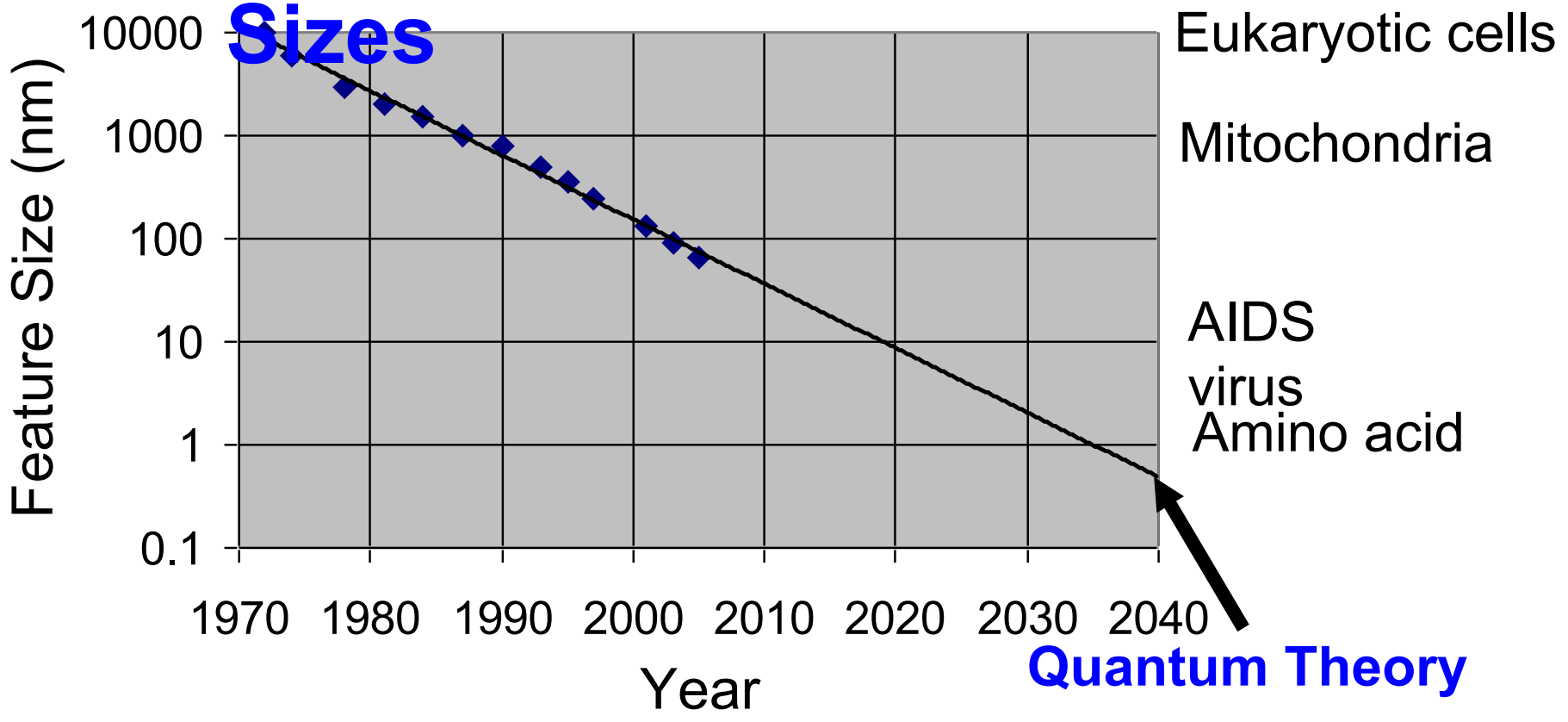
Computation **Physics**

exponential growth of digital computing power combined with the rise of software

elucidation and verification of laws describing how nature operates

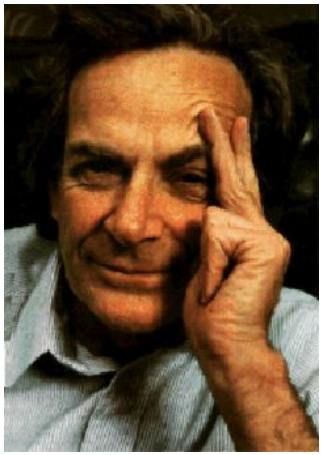
The Future

Computer Chip Feature Sizes



The Future of Computing → Meets → The Past of Physics

Plenty of Time, Right?



Richard Feynman

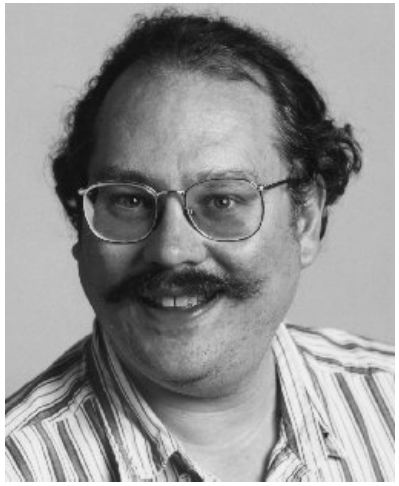
"How can we simulate quantum mechanics?... Can we do it with a new kind of computer…a quantum computer?…It's not a Turing machine but a machine of a different kind." (1981)

"Perhaps…we need a mathematical theory of quantum automata…"
Yu Manin (Radio Moscow, 1980)

1980-1994: Computers operating according to quantum theory?

"A place where really distinguished older physicists went to retire." - Robert Calderbank

A Shocking Discovery



Peter Shor

1994: A quantum computer can efficiently factor integers (breaking an important cryptosystem)

[redacted]-digit integer

1881988129206079638386972394616504398071635633794173827007633564229888597152346654853190606065047430453173880113033967161996923212057340318795506569962213051687593076502570 59

3980750864240649373971255005503864911990643623452526708406385189575946388957261768583317

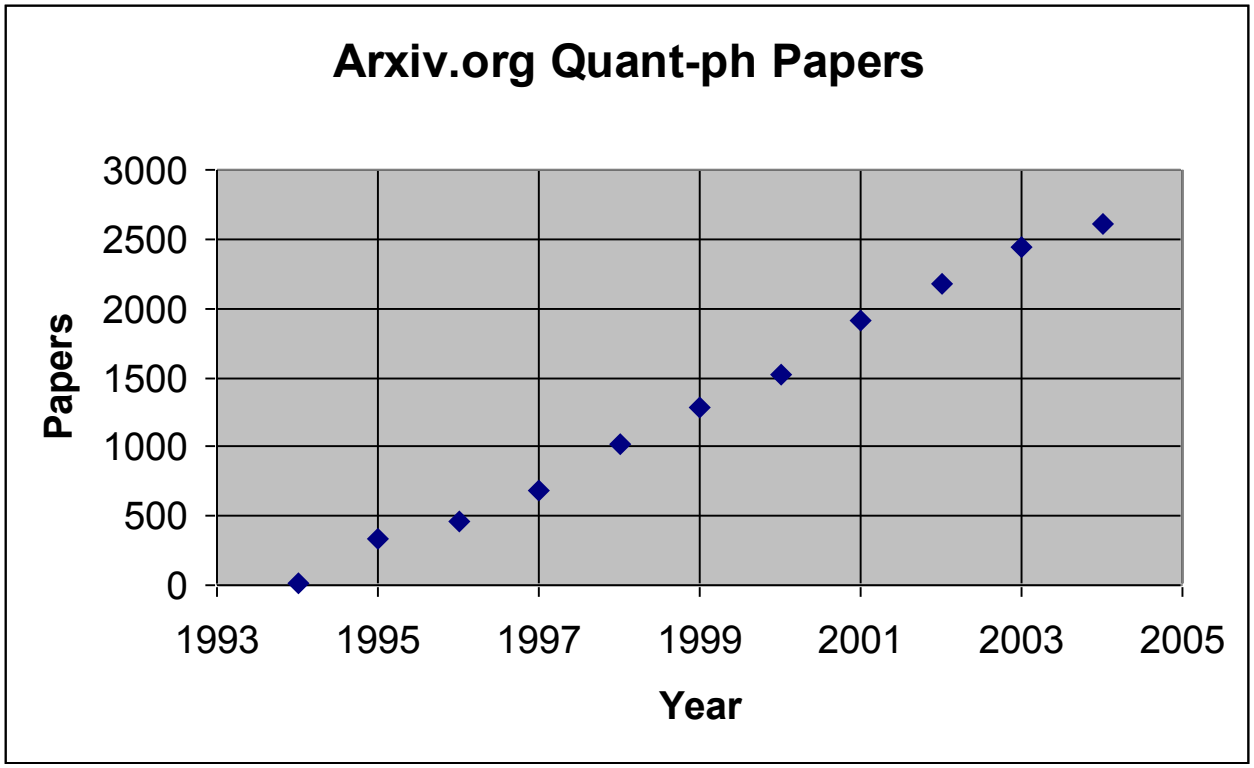
×

4727721461074353025362307197304822463291469530209711645985217113052071125636359039752 7

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's algorithm takes time $O(n^2 \log(n))$

Boom!



Comparison

2004 quant-ph: 2606
2004 hep-th: 3357
(string theory)
2004 cond-mat: 8961

Quantum Information Science

Every single information processing task should be reexamined in light of quantum theory

Quantum Information Science

Physics

Building a Quantum Computer

Physical Quantum Error Correction

Quantum Devices

Quantum Theory

Quantum Cryptography

Quantum Error Correction

Quantum Entanglement

Computer Science

Quantum Algorithms

Quantum Computational Complexity

Quantum Information Theory

Where Do You Want To Go Tomorrow?

This Talk:

What will quantum information science look like in 2020?

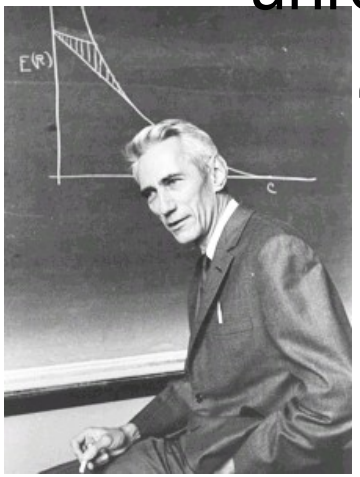
Two and Half Hard Questions:

How will we build a quantum computer?

What gives a quantum computer its power and how can we exploit this to find new quantum algorithms?

To Build a ~~Quantum~~ Computer Classical

Problem (1938): How to build a reliable system from unreliable components?



Claude Shannon

"A mathematical theory of communication" (1948)

How to communicate in the presence of noise

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)



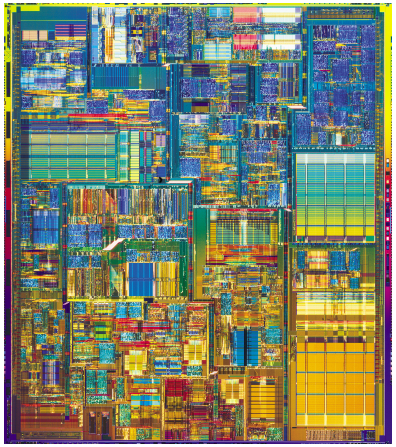
John von Neumann

How to compute when the subsystems of your computer can fail?

No Need for Theory?

Inventions which led to the computer revolution:

Integrated Circuit



Hard Drive



The physics of these devices enacts the methods of the theory (von Neumann and Shannon)!

Two Paths

Hard Drive

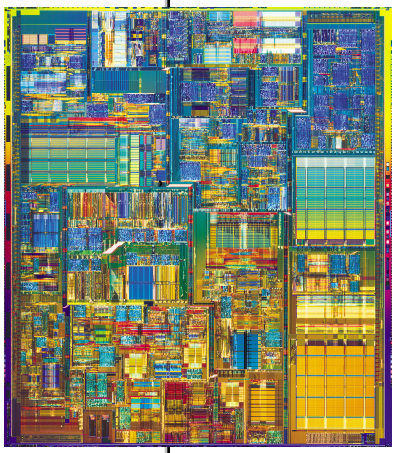


Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerance: macroscopic applied field fixed by local energy minimziation

Integrated Circuit



Coding: majority vote of current

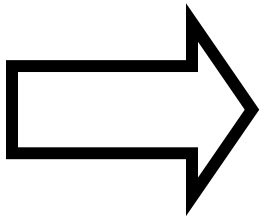
Error correction: amplification

Fault-tolerance: conducting / insulating phase transition

To Build a Quantum Computer

Quantum Computer: Information is now quantum.

probabilistic bit:
p% 0, 1-p% 1



qubit:

$$\alpha|0\rangle + \beta|1\rangle, \alpha, \beta \in \mathbb{C}$$

n qubits:

$$|\psi\rangle \in (\mathbb{C}^2)^n$$

Evolution is now quantum

Schrödinger equation, i.e. unitary evolution

$$|\psi_{out}\rangle = U|\psi_{in}\rangle \quad UU^\dagger = I$$

Measurement is now quantum

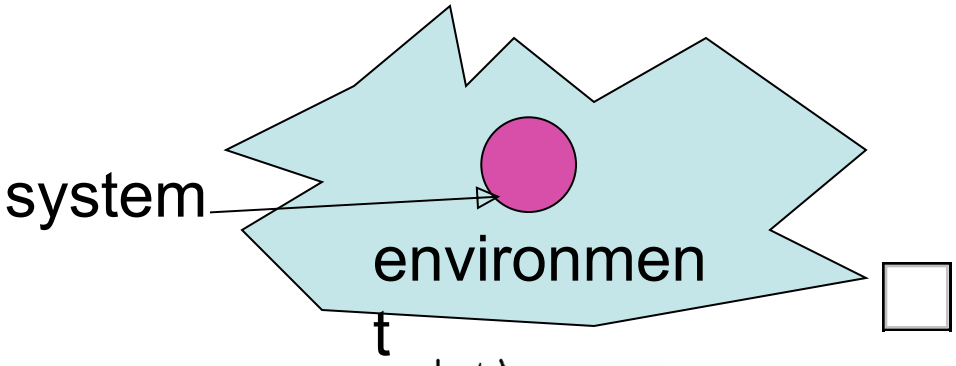
Probability of observing bitstring $|b\rangle$ is $|\langle b|\psi\rangle|^2$

ANALOG? (Compare Digital)

Even Worse!

Problems:

Real quantum systems are open quantum systems



$$|\psi\rangle \rightarrow \rho$$

pure state ⟶ mixed state

$$H = H_{sys} + H_{env} + H_{int}$$

system evolution

environment evolution

system-environment evolution

"system decoheres"

system evolution
is no longer unitary!

Real quantum experiments have finite precision

Perfect unitary evolution is a fiction.
Measurements are faulty. Preparation is imperfect.

Solutions

Observations

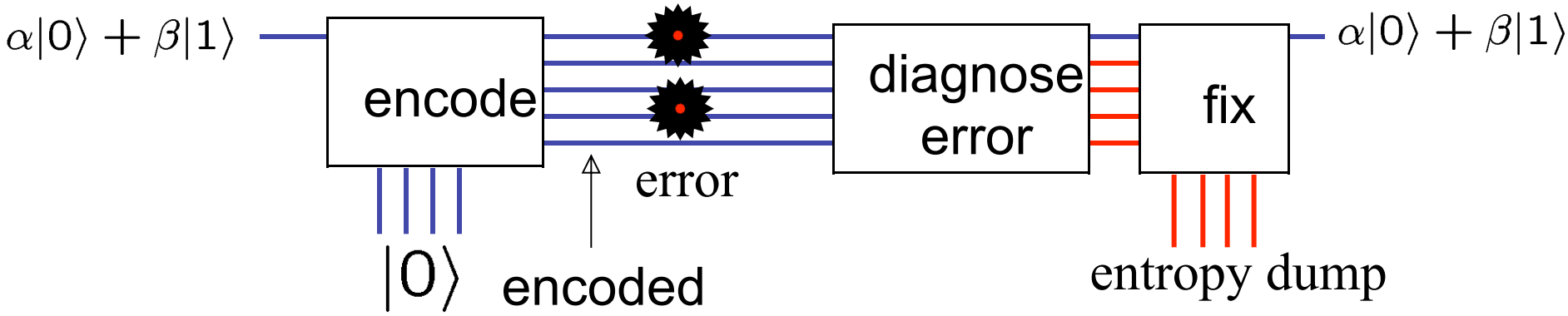
:

Quantum information can be encoded

$\alpha|0_L\rangle + \beta|1_L\rangle, \alpha, \beta \in \mathbb{C}$

$|0_L\rangle, |1_L\rangle$ span a subspace of some Hilbert space $\mathbb{C}^d$

Quantum error correcting codes



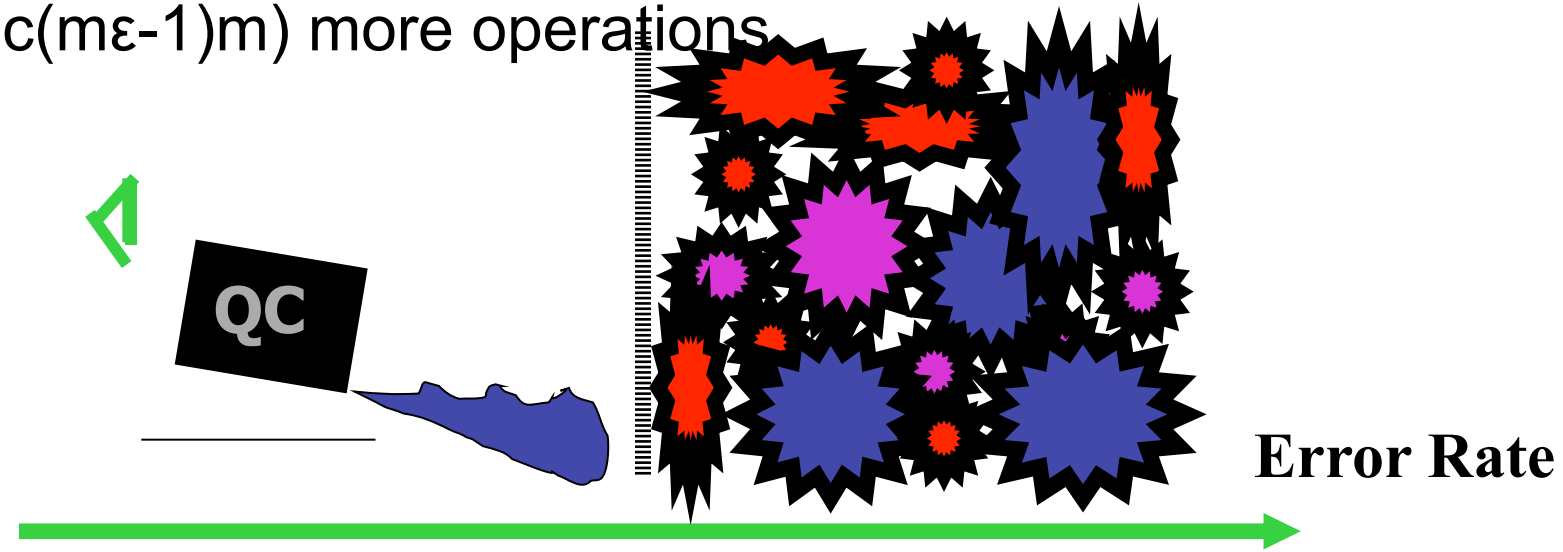
Quantum equivalent to Shannon's noisy coding

Quantum Threshold Theorem

Threshold Theorem:

[Aharanov, Ben-Or, Shor, Zurek, Knill, Laflamme, Kitaev, and Gottesman (95-96)]

If the error rate (decoherence or failure rate per operation) is smaller than some *pthresh*, then quantum computations of length
m can be enacted to accuracy ε at a cost of using a factor O(logc(mε-1)m) more operations.



Quantum computation is digital and is resilient to

Whence Physics?

RANT MODE ON

Today: similar situation to early days of classical computation (threshold theorems but no physics!)

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation

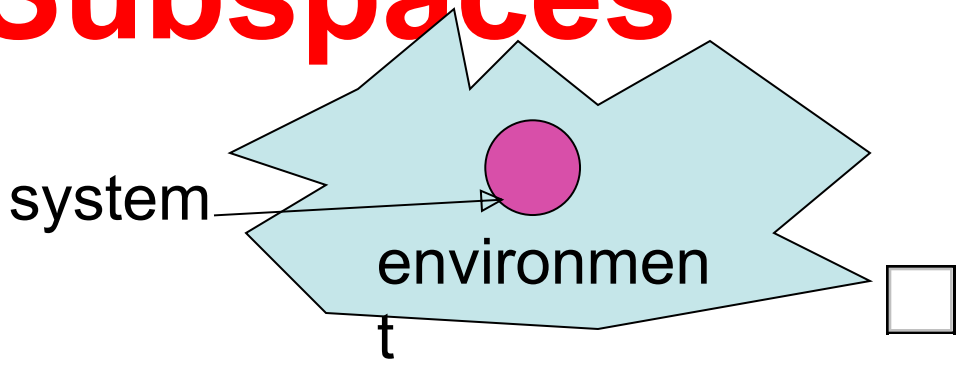
there exists systems whose <u>PHYSICS</u> guarantees their ability to enact robust classical computation.

THE BILLION DOLLAR QUANTUM QUESTION:

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

Decoherence-Free Subspaces

[Zanardi and Rasetti, *Phys. Rev. Lett.* 79, 1953 (1997)]



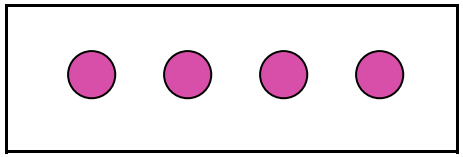
$$H = H_{sys} + H_{env} + H_{int}$$

H_{int}

Return to the physics of decoherence:

Suppose: H_{int} posses a symmetry

Example :



$$H_{int} = \sum_{\alpha} \sum_{i=1}^{4} S_{\alpha}^{(i)} \otimes B_{\alpha}^{(i)}$$

environment cannot distinguish between atoms

$$H_{int} = \sum_{\alpha} \left(\sum_{i=1}^{4} S_{\alpha}^{(i)} \right) \otimes B_{\alpha}$$

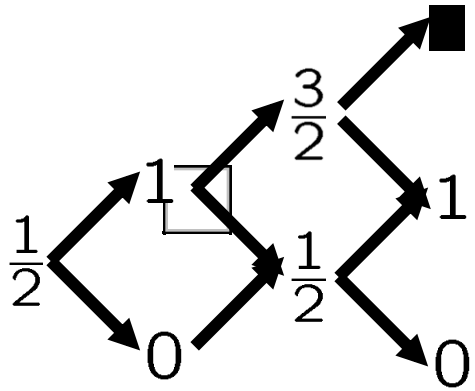
Then: it is often possible to find a subspace of the system which decouples from the environment (decoherence-free)

Collective Resilience

Collective coupling to the environment:

$$H_{int} = \sum_{\alpha} \left(\sum_{i=1}^{4} S_{\alpha}^{(i)} \right) \otimes B_{\alpha} = \sum_{\alpha} S_{\alpha}^{tot} \otimes B_{\alpha}$$

addition of four spin ½ angular momenta



$$S_{\alpha}^{tot} = \sum_{i=1}^{4} S_{\alpha}^{(i)}$$

$$S_{\alpha}^{tot} |S_{tot} = 0\rangle = 0$$

$$H_{int} |S_{tot} = 0\rangle = 0$$

Spin 0 is two-fold degenerate

decoherence does not effect this 2 dimensional subspace

encode a single qubit into this system $|0_L\rangle, |1_L\rangle$

Decoherence-Free Quantum Computation

Symmetry breaking interactions enter at third order (robustness)

[Bacon, Lidar, and Whaley, *Phys. Rev. A* 60, 1944 (1999)]

experimentally verified with photonic qubits

[Altepeter *et al*, *Phys. Rev. Lett.* 92, 14 (2004)]

Alternative symmetric decoherence mechanisms

[Lidar, Bacon, Kempe, and Whaley, *Phys. Rev. A* 63, 022306 (2001)]
[Lidar, Bacon, Kempe, and Whaley, *Phys. Rev. A* 63, 022307 (2001)]

Combining with quantum error correcting codes

[Lidar, Bacon, and Whaley, *Phys. Rev. Lett.* 82, 4556 (1999)]

How to manipulate information encoded into a decohrence-free subspace?

Encoded Quantum Computation

How to compute with encoded quantum information?



$$\alpha|0_L\rangle + \beta|1_L\rangle$$

Want to manipulate encoded operations

- using realistic interactions
- not leave the protected subspace

Exchange: Hamiltonian between i and jth qubits $H_{ij} = J\vec{S}_i \cdot \vec{S}_j$

$\vec{S}_i$ =vector of Pauli spin operators

exchange preserves the decoherence-free subspace

$$[H_{ij}, S_\alpha^{tot}] = 0$$

Universal Quantum Computation

Exchange alone can be used to perform any quantum computation

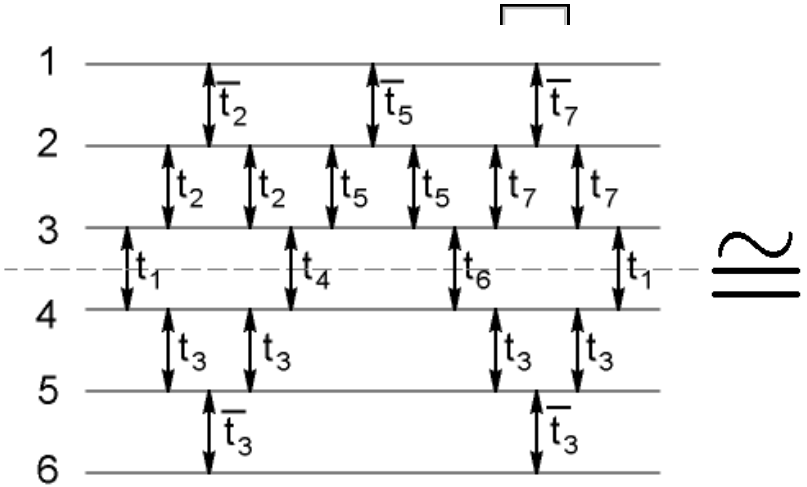
[Bacon, Lidar, and Whaley, *Phys. Rev. Lett.* 85, 1758 (2000)]
[DiVincenzo, Bacon, Kempe, Burkard, and Whaley, *Nature* 408, 339 (2001)]
[Kempe, Bacon, Lidar, and Whaley, *Phys. Rev. A* 63, 042307 (2001)]

encode into decoherence-free subspace (or subsystem)

Controlled Not Gate

1st encoded qubit

2nd encoded qubit



$\simeq$

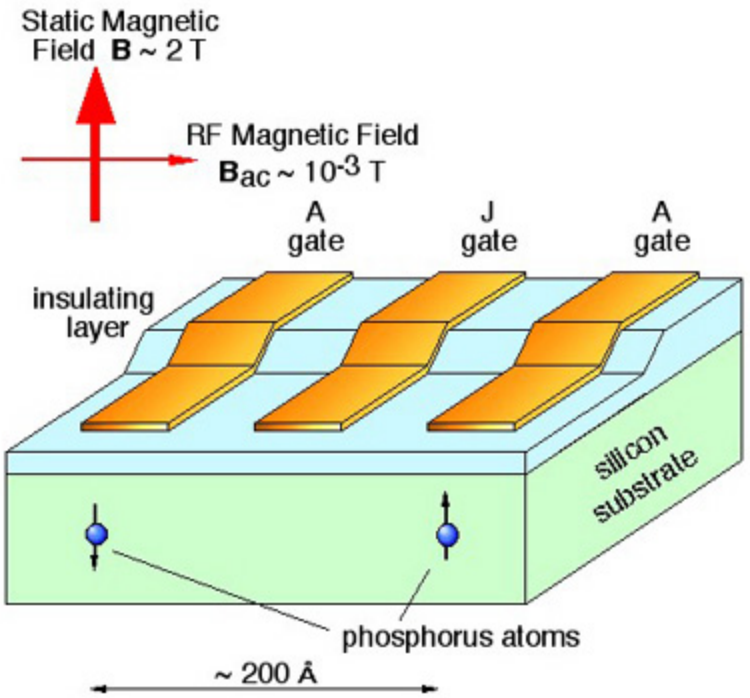
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

t_1=0.410899(2) t_5=0.414720(10)
t_2=0.207110(20) t_6=0.147654(12)
t_3=0.2775258(12) t_7=0.813126(12)
t_4=0.640505(8) $\tan(\pi t)\tan(\pi \bar{t}) = -2$

Encoded Universality

[DiVincenzo, Bacon, Kempe, Burkard, and Whaley, *Nature* 408, 339 (2001)]
[Bacon, Kempe, DiVincenzo, Lidar, and Whaley, *Proc. Int. Conf. on Exp. Imp. of Quant. Comp.* Sydney, Australia (2001)]
[Kempe, Bacon, DiVincenzo, and Whaley, *Quant. Inf. & Comp.* 1, 33 (2001)]

Turn given interactions into a universal quantum computer by using encoded information



Kane Model

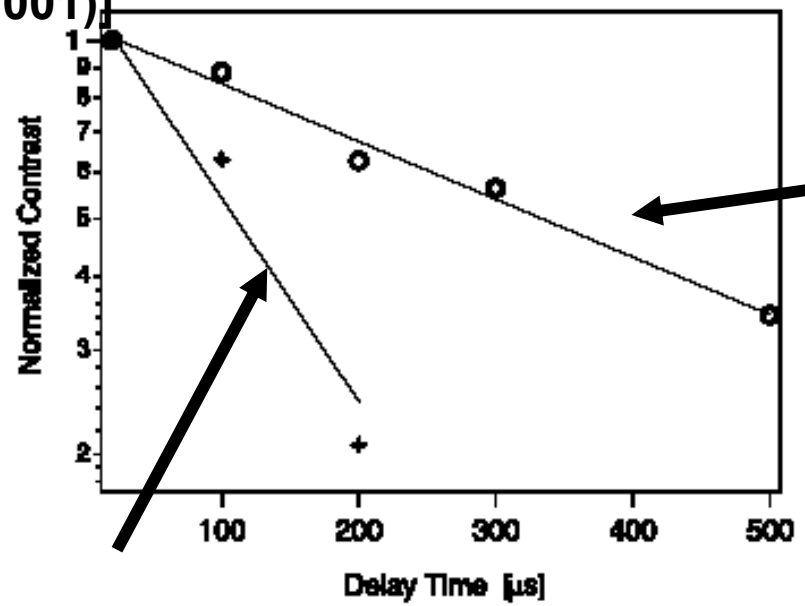
[Kane, Nature 393, 133 (1998)]

~0.1 meV
exchange interaction

single qubit gates not needed

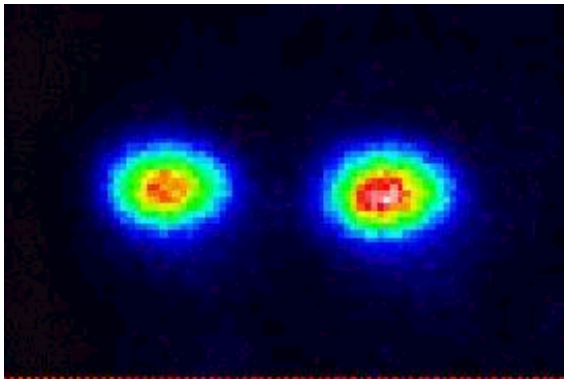
What's Missing?

[Kielpinski *et al* Science 291, 1013 (2001)]



DFS

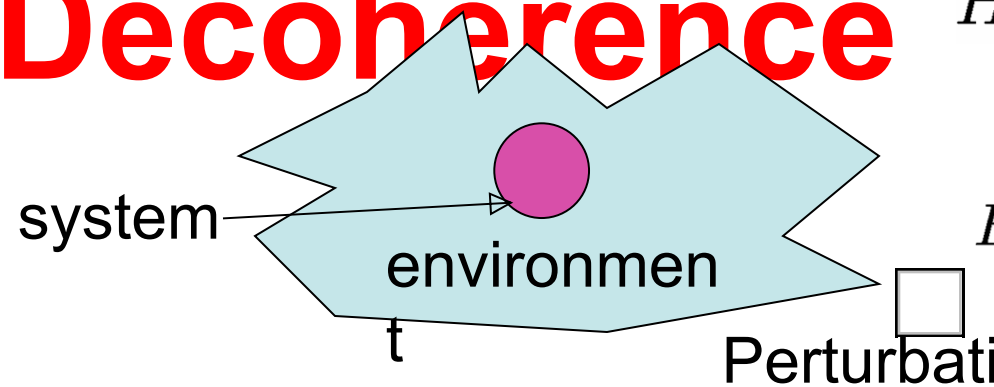
non-DFS



NIST ion trap quantum computing

Physics **Helped,** but we need more help from physics…..

Energetics of Decoherence

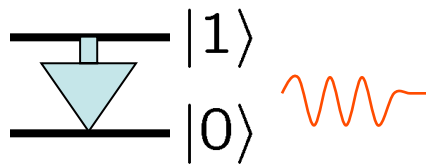


$$H = H_{sys} + H_{int} + H_{env}$$



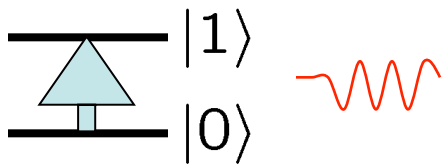
Perturbative regime: $E_{int} \ll E_{sys}, E_{env}$

Perturbative regime: decoherence conserves system and environment energy.



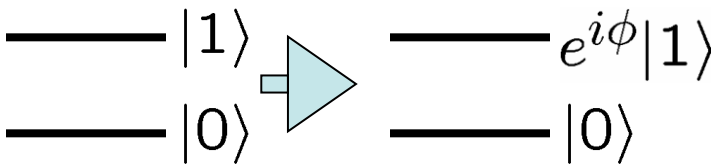
coolin

$\Delta E_{sys} < 0$



heating

$\Delta E_{sys} > 0$



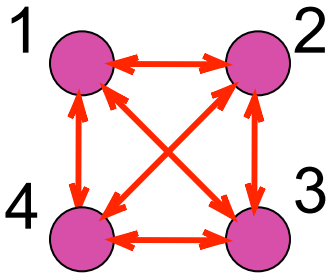
dephasing

$\Delta E_{sys} = 0$

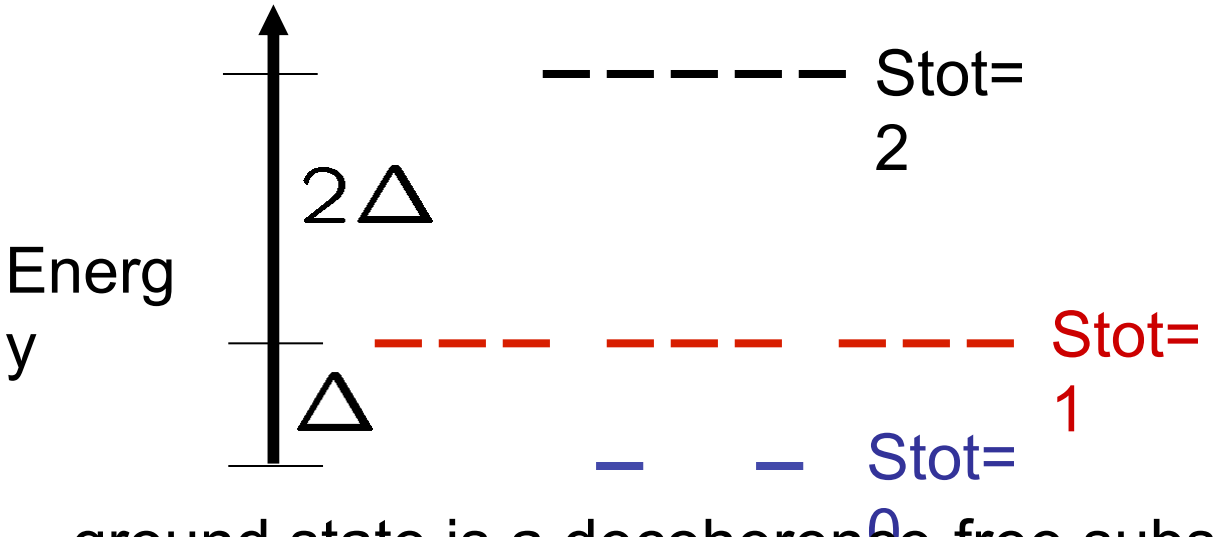
by **cooling environment** we can eliminate heating

Exchange Everywhere

Can we engineer systems which turn dephasing/cooling → heating?



$$H = \Delta \sum_{i>j=1}^{4} \vec{S}_i \cdot \vec{S}_j = \frac{\Delta}{2}\left(S_{tot}^2 - 3I\right)$$



ground state is a decoherence-free subspace

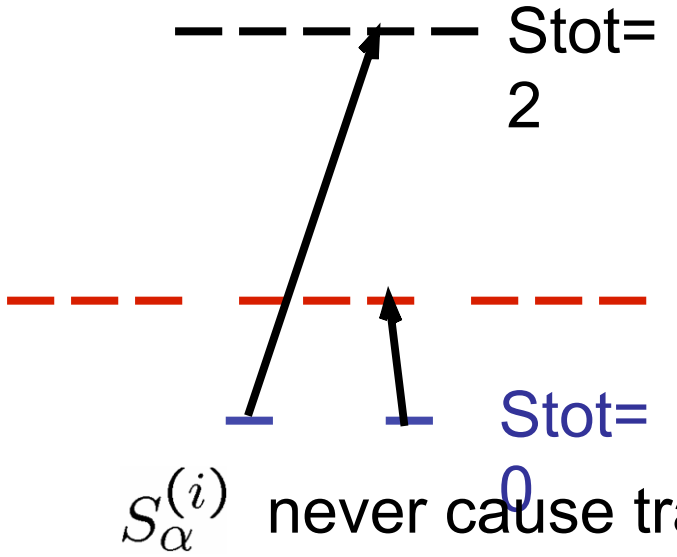
Encode quantum information into ground state

Supercoherence

[Bacon, Brown, and Whaley, *Phys. Rev. Lett.* 87, 247902 (2001)]

Independent coupling of system to environment

$$H_{int} = \sum_{i=1}^{4} S_\alpha^{(i)} \otimes B_\alpha^{(i)}$$



all single qubit interactions on this system are of heating (ground state is quantum error detecting code)

never cause transitions on encoded space

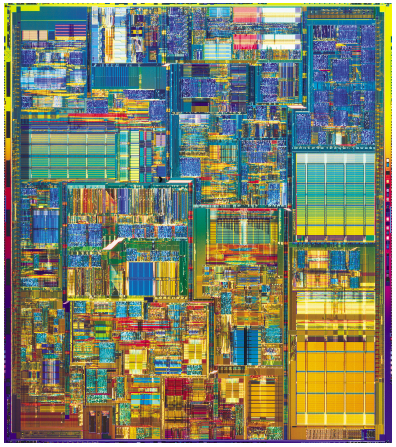
All first order decoherence processes suppressed $kT \ll \Delta$

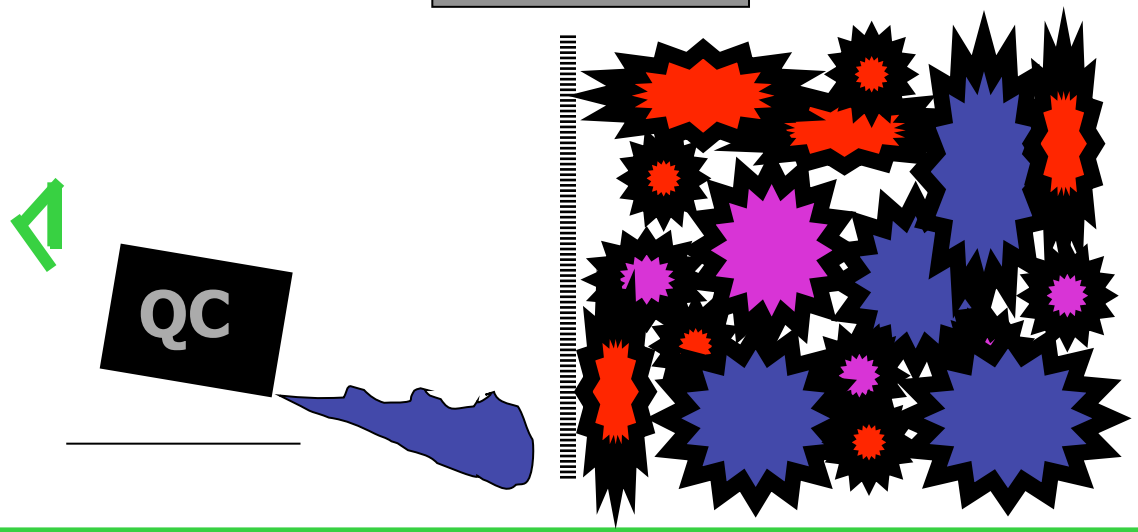
The Path to 2020

Hard Drive

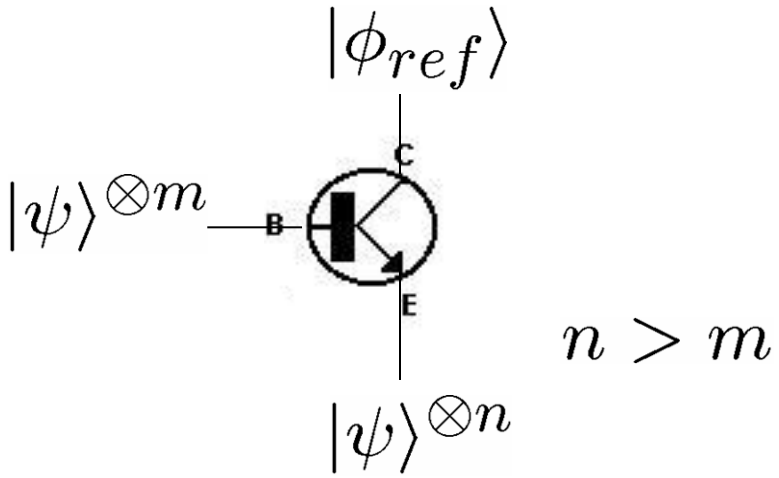


Integrated Circuit





No Quantum Transistor





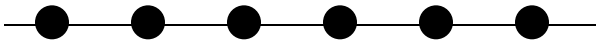
Quantum Cloning Machine

[Wootters and Zurek, *Nature*, 299, 802 (1982)]

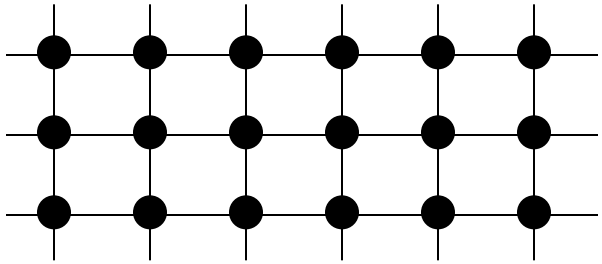
The Physics Guarantee

Classical **Ising model**: $H = -J \sum_{\langle i,j \rangle} Z_i Z_j$

one dimension



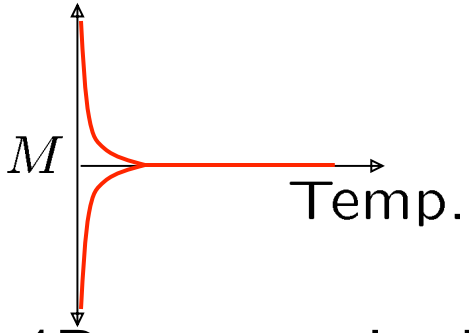
two dimensions



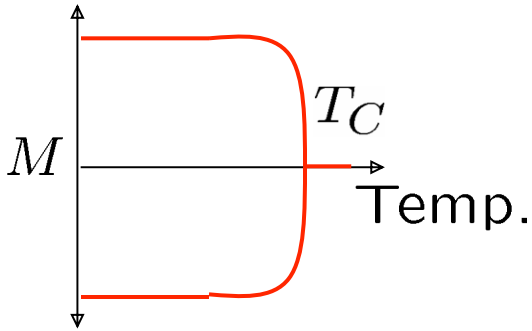
Redundancy code: $|0 \cdots 0\rangle,\ |1 \cdots 1\rangle$

Order parameter: total magnetization $M = \sum_i Z_i$

Information "still there"

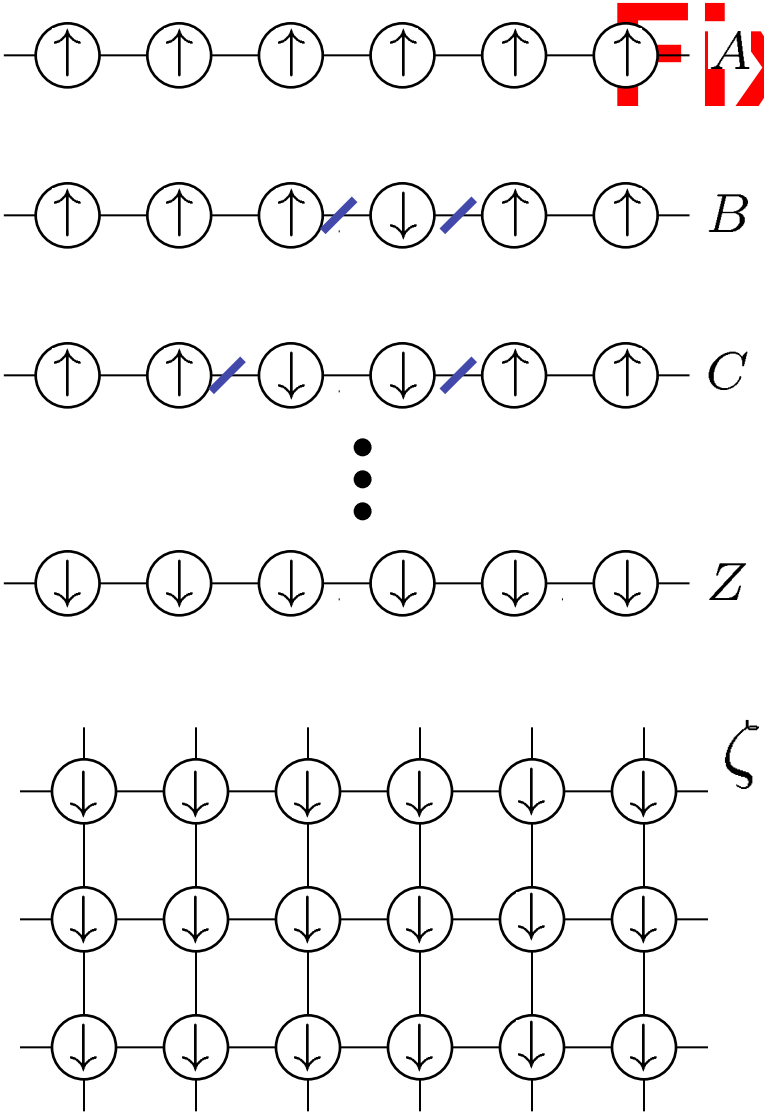


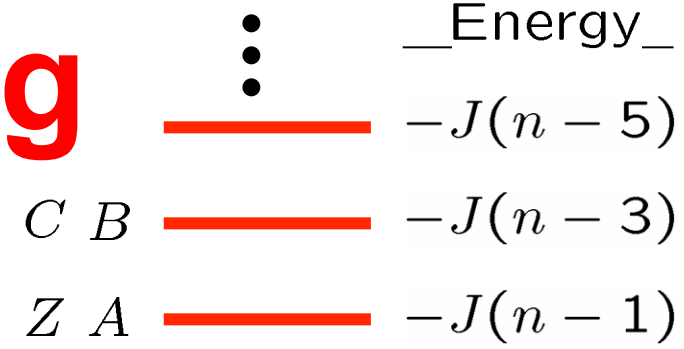
1D memory bad



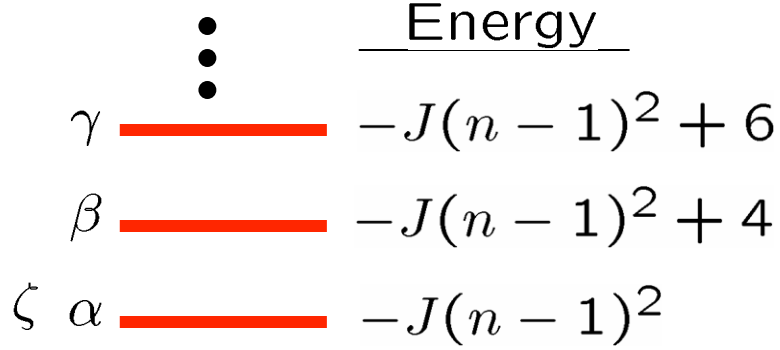
2D memory good

Self Fixing



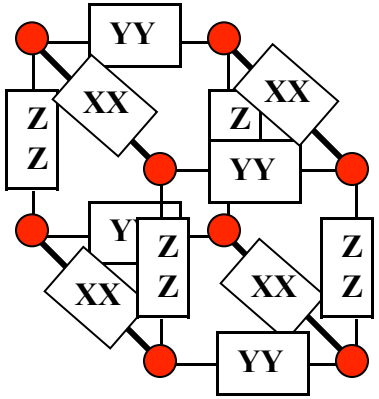


Excitations point-like
disordering entropy >> energy

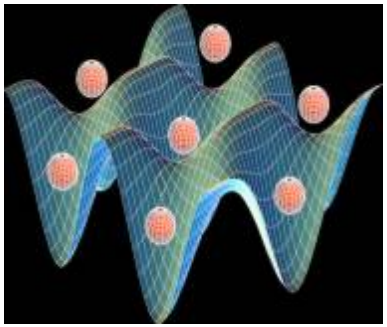


Excitations string-like
Energy grows like perimeter
disordering entropy << energy

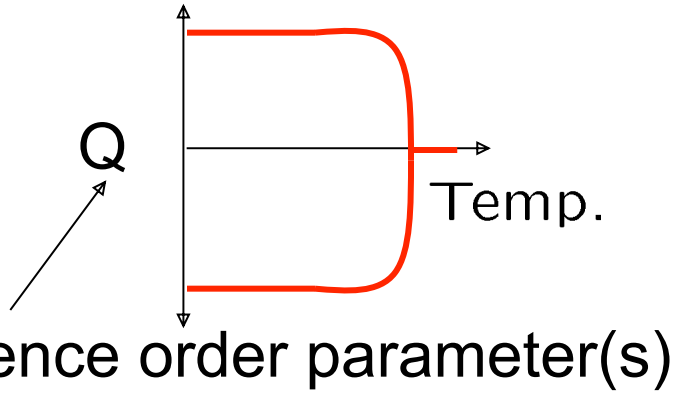
Self-Correcting Quantum Computers



Quantum many-body systems which have excitations which are string-like and are self-correct, but into which we can encode quantum information?



optical lattice



[Bacon, Ph.D. thesis, U.C. Berkeley 2001]
[Bacon, "Quantum Error Correcting Subsystems" in preparation]

Building a Quantum Computer 2020 Edition

traditional approach

control single quantum systems

control two qubit interactions

scale up by adding more and more qubits

pay device complexity cost for quantum error correcting micro architecture

self-correcting approach

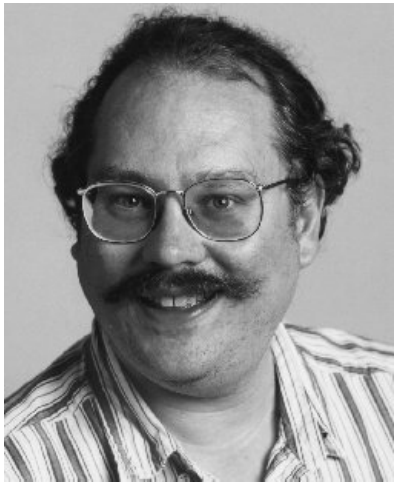
engineer many body systems

robustness guaranteed by the physics of the device

scale down from large systems to smaller

gain exponential number of qubits with increasing fabrication technology

Algorithms!



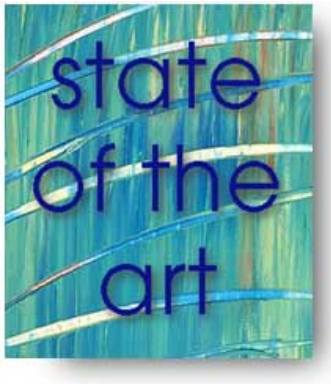
Peter Shor

188198812920607963838697239461650439807
163563379417382700763356422988859715234
665485319060606504743045317388011303396
716199692321205734031879550656996221305
168759307650257059

Factoring is important because efficient factoring algorithms break the most widely used public key cryptosystem (RSA)

What else can a quantum computer efficiently compute?

State of the Art



Quantum Algorithms

- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Various hidden subgroup problems [Long List]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …

Algorithms and Entanglement

Recall: Two subsystem pure states can be separable:

$$|\psi\rangle = |\psi_L\rangle \otimes |\psi_R\rangle$$

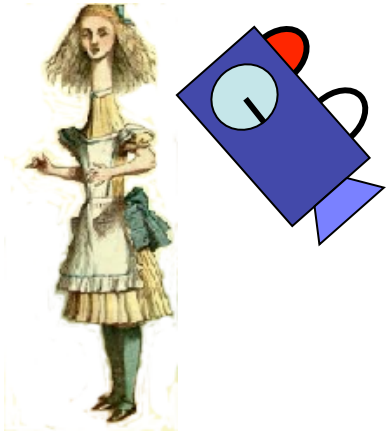
or entangled, i.e.

$$|\psi\rangle = \sum_i \sqrt{p_i}|i_L\rangle \otimes |i_R\rangle$$

Evidence now suggests that entanglement is an essential component of the speedup offered by quantum computers

Entanglement has traditionally only been important for those worrying about the foundations of quantum theory….no more!

Bell's Theorem



ALICE

two quantum subsystems

2. Separate

1. Interact and entangle



BOB

3. Measure subsystem

3. Measure subsystem

Bell's Question: can we explain the measurement outcomes by assuming there were variables (hidden) which were carried by each subsystem?

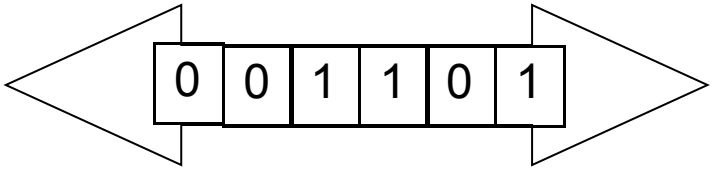
Bell's Answer: not for entangled quantum systems.

Note: correlations from entangled quantum systems cannot be used to communicate between Alice and Bob.

Communication Complexity

How much (many bits) do Alice and Bob need to communicate (worst case) in order to computer a joint function F(a,)B?







ALICE

a

0 1 0 1 ••• 0

?

bits

BOB

b

0 1 1 1 ••• 0

Fxor(a,b)=a1⊕a2 ⊕ …⊕an⊕b1⊕b2 ⊕ …⊕bn

No error: 1 bits

Quantum Communication Complexity



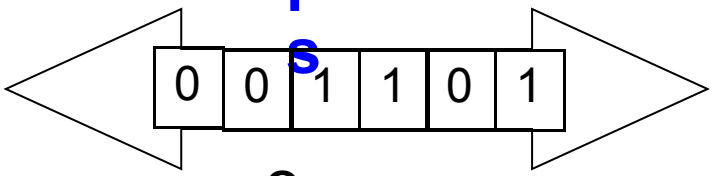
a

ALICE

? entangled bits (ebits)

$$|\psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

plus



?

bits



b

BOB

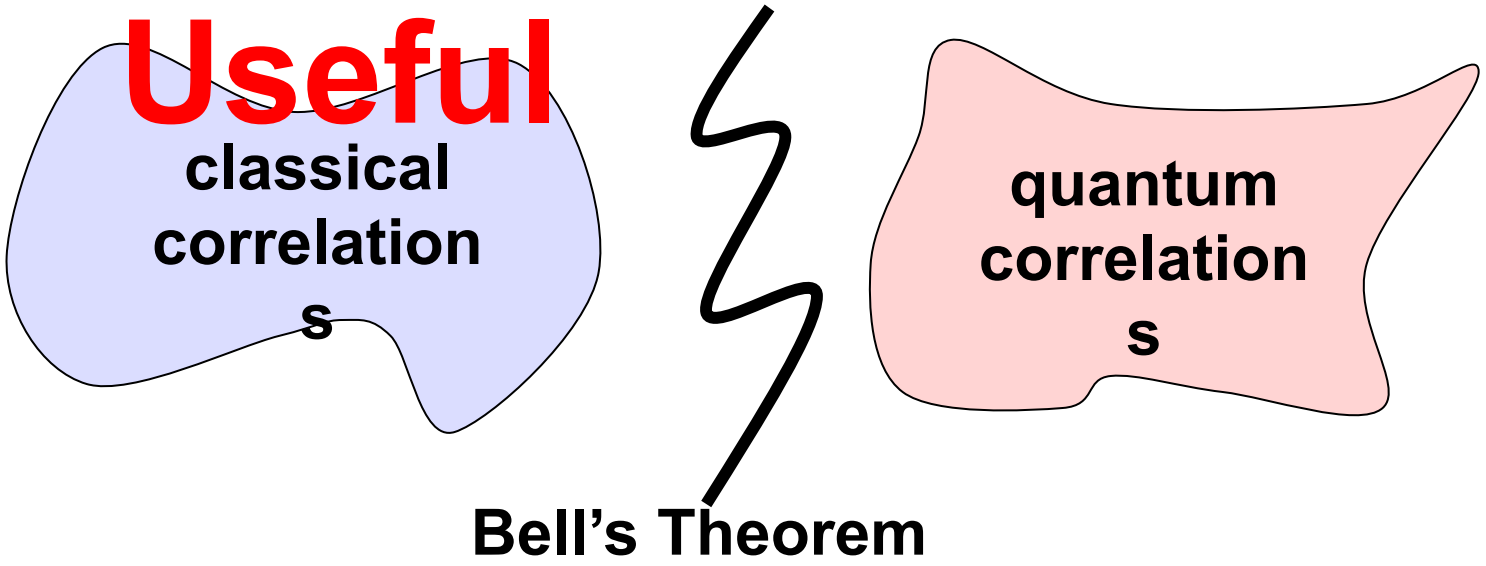


Ran Raz (1999)

There exist functions F(a,b) which require Ω(m1/4 log-1(m)) bits of communication to compute but can be computed using only O(log(m)) bits and ebits [m is the input lengths]

Exponential gain of quantum over classical

Making Bell Useful



Bell's Theorem divides classical from quantum, but does not quantify the power of quantum correlations.

Compare:

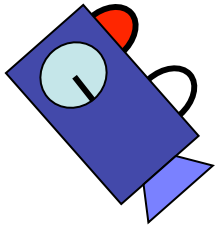
classical correlations plus b bits of communication	to	quantum correlations

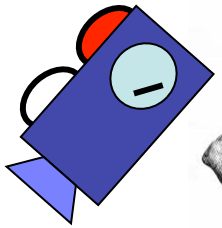
Compar

e

two quantum subsystems









ALICE

3. Choose measurement

4. Get measurement outcome

2. Separate

1. Interact and entangle

3. Choose measurement

4. Get measurement outcome

BOB

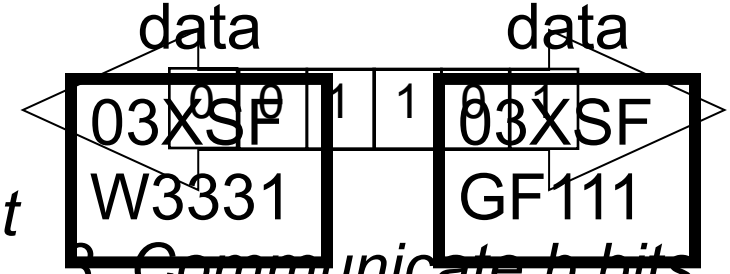


ALICE

2. Choose measurement

4. Produce measurement outcome

1. Share classical correlations



2. Choose measurement

4. Produce measurement outcome



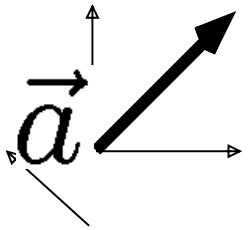
BOB

The Cost of One Ebit



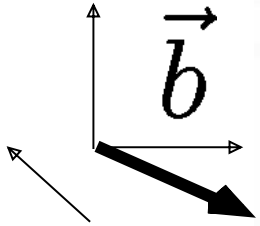
ALICE

measurement direction



$$|\psi_-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

measurement direction





BOB

measurement outcome

$A \in \{+1, -1\}$

measurement outcome

$B \in \{+1, -1\}$

Quantum correlations:

$$Pr(A, B|\vec{a}, \vec{b}) = \frac{1}{4}\left[1 - \alpha\beta\vec{a}\cdot\vec{b}\right]$$

How to produce these correlations without sending infinite string describing measurement directions?

Surpris

[Bacon, and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
[Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]

The **Bell Experiment** can be simulating using

classical correlations

plus

one bit of communication

1. Alice and Bob share two sets of perfectly classically correlated unit vectors randomly distributed over all directions: $\vec{\lambda}_1, \vec{\lambda}_2$

2. Alice outputs $A = -\mathrm{sgn}\left(\vec{a} \cdot \vec{\lambda_2}\right)$

3. Alice sends a bit c to Bob $\quad c = \mathrm{sgn}\left(\vec{a} \cdot \vec{\lambda}_1\right) \mathrm{sgn}\left(\vec{a} \cdot \vec{\lambda_2}\right)$

4. Bob outputs $B = \mathrm{sgn}\left(\vec{b} \cdot \left(\vec{\lambda_1} + c\vec{\lambda_2}\right)\right)$

Finiteness Conjecture

Conjecture: all multipartite entangled systems can be exactly simulated by augmenting a classical correlations with an amount of communication polynomial in the product of the dimensions of the entangled subsystems.

- projective measurements on two entangled qubits
  [Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]
  [Bacon and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
- all bipartite two outcome correlation observables
  [Bacon and Toner, in preparation (2005)]
- Pauli measurements on a n-party GHZ states
  [Tessier, Caves, Deutsch, Eastin, Bacon, quant-ph/ 0407133 (2005)]
- average case with projective measurements on n ebits
  [Massar, Bacon, Cerf, Cleve, *Phys. Rev. A* 63, 052305 (2001)]
- ….

Back To Efficiency



Guifre Vidal

Small entanglement implies there is an efficient classical simulation.

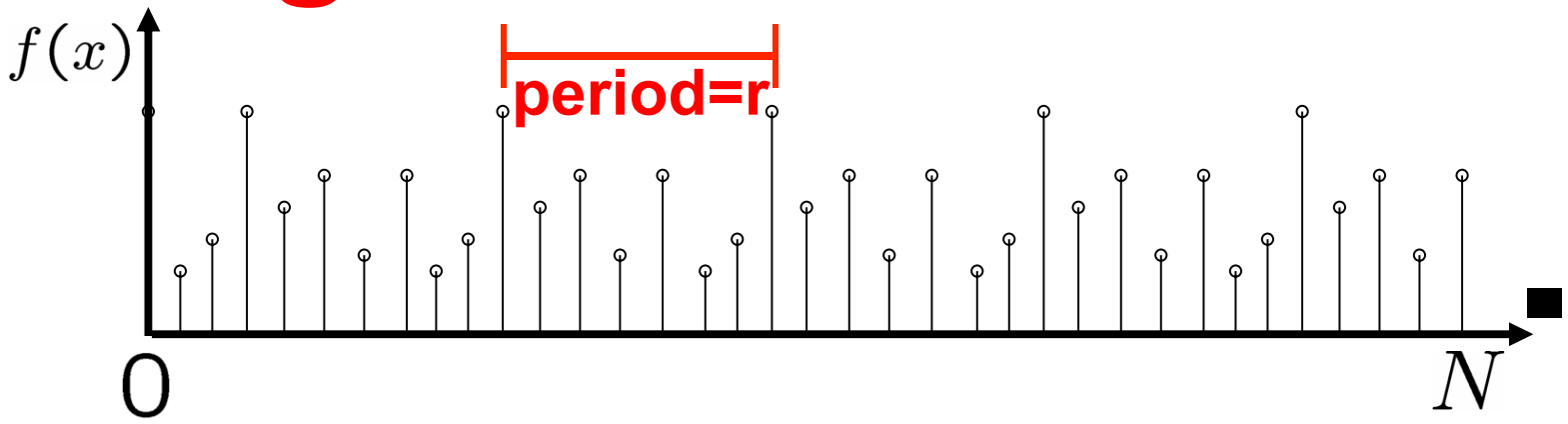
[G. Vidal, Phys. Rev. Lett. 93, 040502 (2004)]

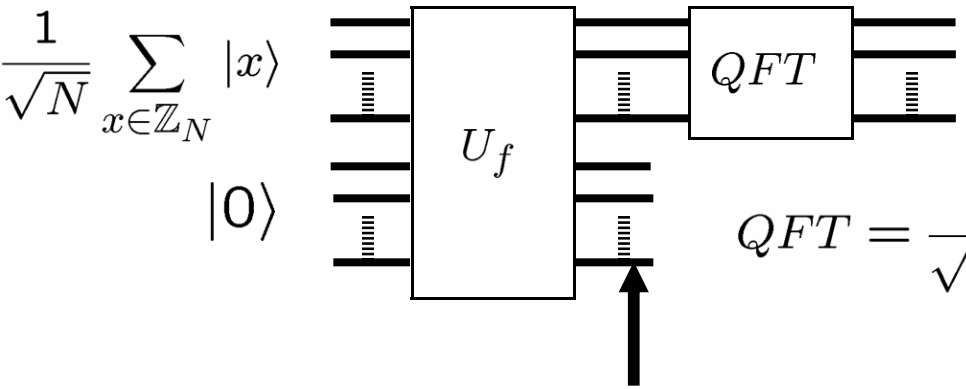
Entanglement is necessary for a quantum speedup.

How does this influence how we use quantum devices?

Factoring and Entanglement

Factoring can be reduced to period finding.





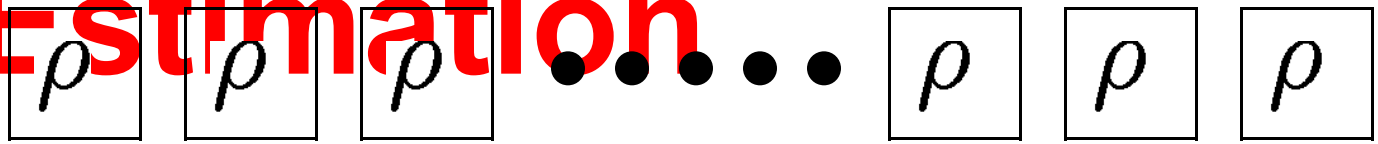
measure and use to find r

$$QFT = \frac{1}{\sqrt{2^n}} \sum_{x,p=0}^{2^n-1} \exp\left(\frac{2\pi i p x}{2^n}\right) |p\rangle\langle x|$$

$$\frac{1}{\sqrt{N}} \sum_{x \in \mathbb{Z}_N} |x\rangle \otimes |f(x)\rangle$$

highly entangled basis

Spectrum Estimation



n copies of a d-dimensional unknown mixed state

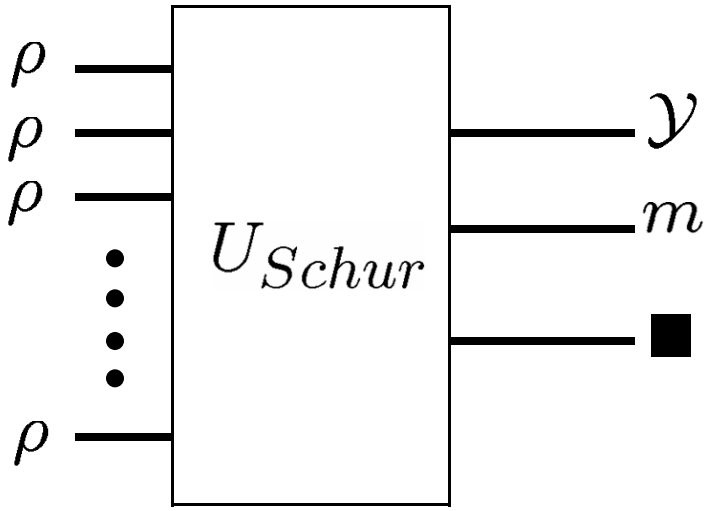
$$\rho = \sum_i p_i |\phi_i\rangle\langle\phi_i| \longleftarrow \text{we don't know } |\phi_i\rangle$$

Best method to estimate the *pi*?

Try: Separate system measurements? (state tomography)

No: optimal measurement is to project onto entangled states across all n subsystems

Schur Transform

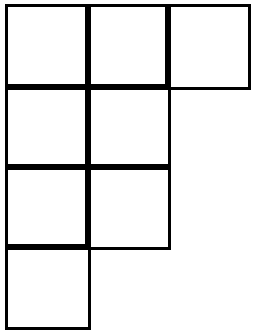


entangled basis for best estimating the spectrum

$$U(d)^{\otimes n} = \bigoplus_{\mathcal{Y}} I_{d_{\mathcal{Y}}} \otimes U_{\mathcal{Y}}(d)$$

Young diagram of size *n*

Measuring $\mathcal{Y}$ estimates spectrum

$\mathcal{Y}$	$\rho = \sum_i p_i \lvert\phi_i\rangle\langle\phi_i\rvert$
n=8  *r1*= 3 *r2*= 2 *r3*= 2 *r4*= 1	*p1*≈3/8 *p2*≈1/4 *p3*≈1/4 *p4*≈1/8

How to efficiently implement as quantum circuit?

One Fell Swoop

Efficient quantum circuit constructions:

[Bacon, Harrow, Chuang, quant-ph/0407082, (2004)]

yields efficient quantum circuits for….

Spectrum Estimation

[Keyl and Werner, *Phys. Rev. A* 64, 052311 (2001)]

Quantum Hypothesis Testing

[M. Hayashi, *J. Phys. A* 35, 10759 (2002)]

Universal Quantum Source Coding

[M. Hayashi and K. Matsumoto, *Quant. Inf. Comp.* 2, 519 (2002)]

[M. Hayashi and K. Matsumoto, *Phys. Rev. A* 66 022311 (2003)]

Universal Entanglement Concentration

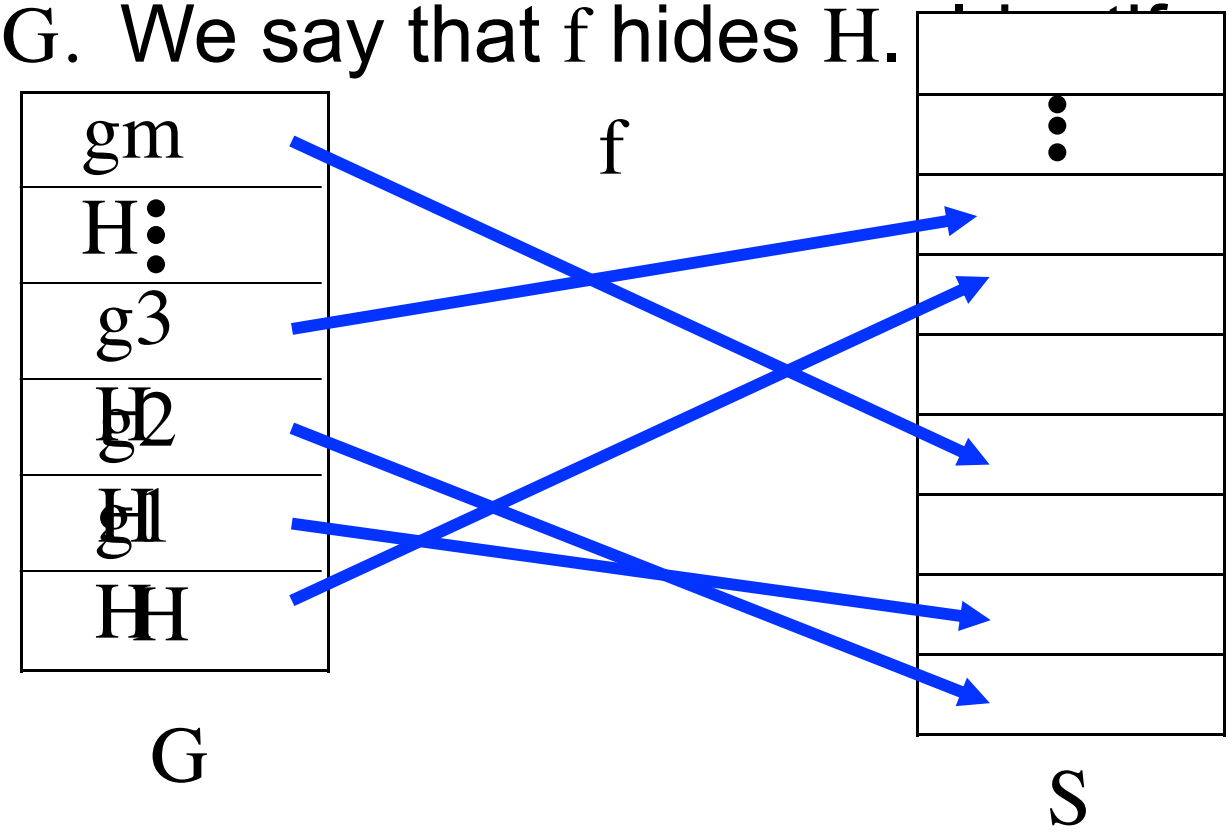
[M. Hayashi and K. Matsumoto, quant-ph/ 0209030 (2002)]

Communication Without a Shared Reference Frame

[S. Bartlett, *et al*, *Phys. Rev. Lett.* 91 027901 (2003)]

Future Algorithms

Problem: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that is constant and distinct on left cosets of H in G. We say that f hides H. Identify H.



Hidden Subgroup Problems

Shor's algorithm is an example of a HSP.

Efficient algorithms for the HSP over

the Symmetric group → an efficient algorithm for graph isomorphism

the Dihedral group → an efficient algorithm for poly(d)–unique shortest vector in a lattice

Phase Transition

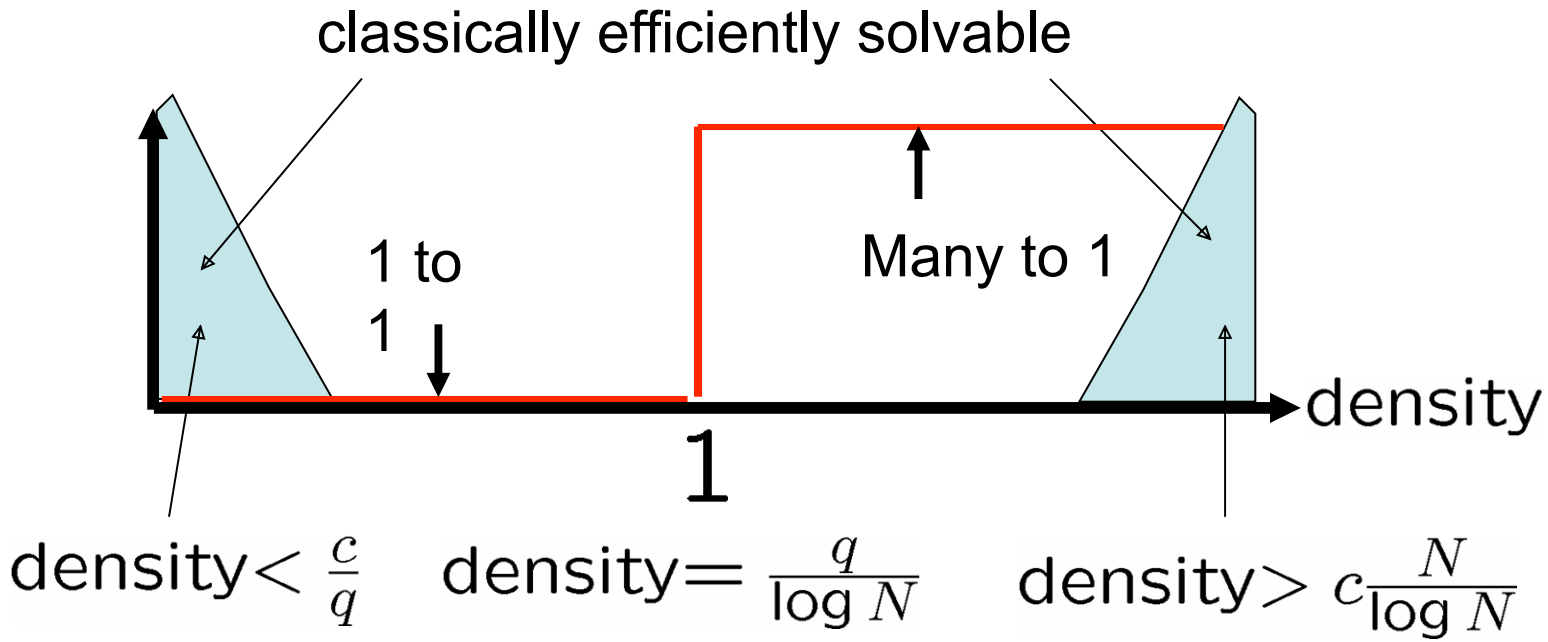
Subset Sum Problem [NP-complete]

Given: $S = \{J_1, J_2, \ldots, J_q\}, J_i \in \mathbb{Z}_N$

$t \in \mathbb{Z}_N$

Problem: Is there a subset of *S* which sums mod N to *t*?

$$f(b) = b_1 J_1 + b_2 J_2 + \ldots + b_q J_q \bmod N, \; b_i \in \mathbb{Z}_2$$



Dihedral HSP

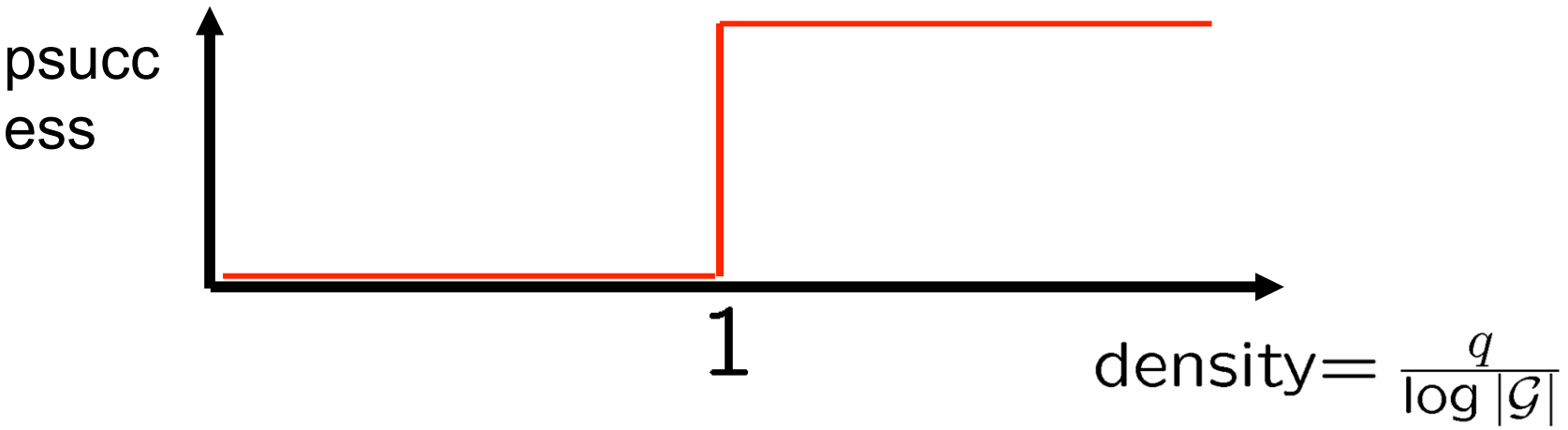
[Bacon, Childs, van Dam, quant-ph/0501044 , (2005)]

Optimal measurement for the Dihedral Hidden Subgroup Problem

$\rho_{\mathcal{H}}$ $\rho_{\mathcal{H}}$ $\rho_{\mathcal{H}}$ ● ● ● ● ● $\rho_{\mathcal{H}}$ $\rho_{\mathcal{H}}$ $\rho_{\mathcal{H}}$

q copies of hidden subgroup states

Optimal measurement is entangled measurement and…



202

How to build a quantum computer? Use physics:

Attempt 1: **Decoherence-free subspaces**
passive
(discover encoded universality)

Attempt 2: **Supercoherent Quantum Bits**
use spectrum of few qubit systems

Attempt 3: **Self-correcting Quantum Computers**
jumpstart the quantum computer revolution?

Quantum algorithms? Entanglement matters:

Attempt 1: **The Cost of Quantum Correlations**
1 bit + classical = ebit experiment

Attempt 2: **Schur Transform for Quantum Tasks**
efficient quantum circuits

Attempt 3: **Hidden Subgroup Problems**
phase transitions and entangled measurements

Collaborators



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IBM T.J. Watson
David Divencenzo
Guido Burkard

U.C. Berkeley
K. Birgitta Whaley

Stanford
Xinlan Zhou

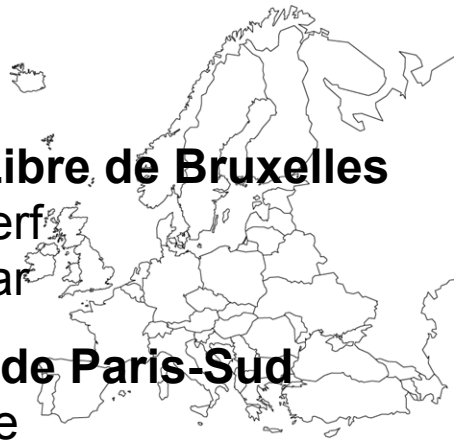
U.C. Santa Barbara
Wim van Dam

Univ. of New Mexico
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Ivan Deutsch
Brian Eastin
Tracey Tessier

Caltech
Andrew Childs
Debbie Leung
Benjamin Toner



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